
Mohammed Aayan Pathan

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Dubai, UAE

Summary

Full-stack developer and computer scientist with a focus on applied AI, data-driven web apps, and impactful user experiences. Proven track record of building real-world projects with modern technologies and a strong interest in scalable AI systems. Comfortable working in small engineering teams, using Git workflows, code reviews, iterative development cycles and Agile sprints.

Education

A-Levels

GEMS Founders School - Al Barsha • Dubai, UAE

06/2026

A-Levels: Achieved: A* in Mathematics, A in AS Physics and AS Further Math (*predicted: A* in Further Math, A in CS and Physics*)

Subjects studied include: **Mathematics, Further Mathematics, Computer Science and Physics.**

GCSE/iGCSE

GEMS Founders School - Al Barsha • Dubai, UAE

06/2024

GCSEs: Achieved grades 999988877, equivalent to A*A*A*A*A*A*A*AA.

Subjects studied include:

Mathematics, Computer Science, Chemistry, Psychology, Physics, Business Studies, English Literature and Language, Further Pure Mathematics

Work and Volunteering Experience

Junior Engineering Associate - Full-Stack Web Developer

VOXA (voxalab) • Dubai, UAE (remote)

10/2025 - 04/2026

- Refactored the Voxa MVP codebase into version 1.0, modernized architecture, with redefined user stories and responsive design.
- Co-developed the Voxa Demo app for VCs and potential users for the Lisbon WebSummit 2025 event, using React (Vite) and Node.js; responsible for version control and deployment workflow.
- Optimized UX and UI design alongside colleagues, providing assistance and insight with Figma designs to reduce navigation friction.
- Implemented Firebase Functions + App Check to automatically flag and successfully remove 100% of harmful content posted on the community using AI-based moderation, improving platform safety for the MVP.
- Reworked Google Analytics integrations, enabling more precise tracking of user-interaction data and data-driven UX iterations.
- Assisted in onboarding 10+ developer interns: authored a 15+ page thorough developer wiki document, provided code reviews, and ensured Git workflow standards.
- Migrated version 1.0 from Firebase to Supabase for authentication and data storage for scalability and reducing technical debt for the future.

Product and App Development Intern

VOXA (voxalab) • Dubai, UAE (remote)

07/2025

- Developed a full-stack voice training app using Next.js and Firebase, with 5+ **core** features.

- Specialized in frontend design using TailwindCSS to build responsive layouts.
- Implemented 2 APIs: AssemblyAI APIs for live transcription and OpenAI GPT for speech feedback.
- Added user authentication with Firebase Auth, Firebase Storage and Firestore DB to post/delete/bookmark posts and replies, create and modify accounts and store activity and game data.
- Managed GitHub repository and led deployment via Vercel.
- Collaborated with a team of 3 to ensure seamless feature integration and development cycles.
- Conducted user testing (over 30 testers) to gather feedback for iterative improvements.
- Demonstrated proficiency in prototyping UI designs using Figma.

GCSE Intervention Volunteer Tutor

GEMS Founders School - Al Barsha • Dubai, UAE

09/2024 - 04/2025

- Co-taught twice-weekly GCSE Intervention sessions during my free lessons in school, alongside the Head of Computer Science.
- Marked and reviewed dozens of students topic tests and mock examinations, including theory and programming papers for both Year 11 and Year 12.
- Most students passed with some students attaining top grades, and the opportunity to pursue the subject at A-Level.
- Awarded Certificate of Appreciation in Computer Science for my contributions, being the only one in my Year 12 cohort to receive this award.

Student Intern - Team Leader

8BillionIdeas • Dubai, UAE

06/2024

- Led a team in developing and executing a comprehensive marketing strategy for the rebranding of a hypothetical company.
- Collaborated with team members to design wireframes and interactive prototypes for the website redesign.
- Applied user-centric UI/UX principles to improve website functionality, accessibility, and overall experience.
- Guided the team through the full rebranding lifecycle, fostering collaboration, creativity, and efficient workflow.
- Presented design concepts to mock stakeholders, incorporating feedback to refine deliverables and strengthen pitching and communication skills.

Projects

FormulaMetric - formulametric.com

React, Node.js, MySQL, Express, TailwindCSS, Vercel, Render, Aiven

- Designed full-stack F1 analytics platform using React, Node.js, Express, and MySQL with normalized schema, foreign keys, and indexed lookups.
- Created RESTful API with season, driver, and race filters, including pagination and query parameters.
- Implemented data ingestion from the Jolpica API, detecting missing race data and applying a recursive exponential backoff retry algorithm to avoid rate limits.
- Built regression, correlation, and sigmoid-based performance metrics for multi-season comparisons.
- Developed driver/team assignment logic including support for rotating lineups across seasons.
- Utilized a random-pivot recursive Quicksort algorithm utility to facilitate the different metric factors that users can sort drivers by.
- Designed a top-k min-heap (stored as a binary tree with priority queue operations) to efficiently find the top 3 drivers in a season in linear time complexity.
- Built interactive UI with MUI X gauges, charts, and responsive layouts, supporting comparison of up to 3 drivers.
- Implemented loading and fallback states to ensure smooth UX across data-intensive views.
- Deployed the frontend via Vercel and the backend via Render and Aiven and connected them together to enable successful deployment.

Voxa Voice App - voxa.club

Next.js, TypeScript, TailwindCSS, Supabase (migrated from Firebase), PostgreSQL, Google Analytics, Vercel, AssemblyAI, OpenAI

- Built real-time audio streaming pipeline: microphone capture, downsampling, base64 encoding, chunked audio frames, and continuous WebSocket transmission to AssemblyAI's API.

- Integrated live transcription into React state, handling partial and final segments with smooth UI updates and clean stream termination.
- Implemented Supabase with PostgreSQL for authentication, database and file storage.
- Rebuilt large portions of the app UI using TailwindCSS and shadcn/ui, replacing legacy components for improved performance and accessibility.
- Managed GitHub repo, Vercel deployments, environment configs, and multi-developer version control workflow.
- Integrated Google Analytics to track user behaviour and feature usage.

TLSearch – NASA Space Biology Search Engine – tlsearch.org

Next.js, FastAPI, Python, TailwindCSS, Vercel, Render

- Built a semantic search tool for NASA space-biology datasets, converting user queries and document metadata into vector embeddings and ranking them using cosine similarity.
- Implemented FastAPI endpoints that accept search queries, compute similarity scores in Python, and return ranked results as JSON.
- Loaded NASA datasets into Python and processed them in-memory for fast experimentation and retrieval.
- Developed a Next.js frontend that sends search requests to the FastAPI backend and displays results with a responsive TailwindCSS interface and SEO-ready metadata.
- Created a fully functional end-to-end pipeline integrating a vector-based search algorithm with a modern web UI.
- Deployed the frontend via Vercel and the backend via Render and connected them together to enable successful deployment.

Portfolio Website – aayanpathan.com

React, TypeScript, TailwindCSS

- Built a personal portfolio website using React and TypeScript with a clean, responsive layout.
 - Implemented animated backgrounds using Vanta.js to create a dynamic visual experience.
 - Added a custom loading sequence and smooth UI transitions with Framer Motion.
 - Integrated EmailJS to enable direct contact form submissions.
 - Designed all components with TailwindCSS for consistent styling and fast load performance across devices.
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Skills

NextJS, TypeScript, TailwindCSS, Node.js, Express.js, Firebase, MySQL, JavaScript, Python, Claude Code, GitHub, Vercel, Slack, Figma, Supabase, English: IELTS Band 8.5 – C2 Proficiency

Awards, Honors and Certificates

- Star Associate of the Year – VOXA
- ICSC Bronze Finalist w/honors (top 8% internationally)
- Best Student – Year 12 Computer Science Cohort (2024–2025)
- 2x UKMT Bronze
- Claude Code in Action – Anthropic Academy